EXPERIMENT 7

AIM: To apply navigation, routing and gestures in a Flutter application.

OBJECTIVE: To implement navigation between screens using Navigator, organize navigation with named routes, and handle tap, double-tap and long-press gestures using GestureDetector.

COURSE OUTCOME: CO 2 - Design and implement user interfaces using Flutter widgets, layouts, and animations, ensuring responsiveness and interactivity.

THEORY

Flutter manages screen transitions with the Navigator widget. Navigator.push() places a new route on the navigation stack, while Navigator.pop() removes the current route and returns to the previous screen. MaterialPageRoute provides a basic route when a screen is opened directly.

Named routes improve organization by assigning route names in MaterialApp.routes. A screen can then be opened with Navigator.pushNamed(context, '/second'), which keeps navigation definitions centralized and supports a scalable project structure.

GestureDetector makes widgets interactive without changing their appearance. Its callbacks include onTap, onDoubleTap and onLongPress. In this experiment, gestures display SnackBar feedback and update the color of an AnimatedContainer. Navigation and gestures are combined by making the Flutter logo open the second screen through a named route.

SOFTWARE REQUIREMENTS

- Flutter SDK
- Dart SDK
- Android Studio or Visual Studio Code
- Android emulator, physical device, Chrome or Edge

CODE

main dart

```
import 'package:flutter/material.dart';

import 'home_screen.dart';
import 'second_screen.dart';

void main() {
  runApp(const NavigationGestureApp());
}

class NavigationGestureApp extends StatelessWidget {
  const NavigationGestureApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Navigation and Gestures',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFF3157D5),
        ),
        useMaterial3: true,
      ),
      initialRoute: '/',
      routes: {
```

```
        '/': (context) => const HomeScreen(),
        '/second': (context) => const SecondScreen(
              openedWith: 'Navigator.pushNamed()',
            ),
      },
    );
  }
}
```

home screen dart

```
import 'package:flutter/material.dart';

import 'second_screen.dart';

class HomeScreen extends StatefulWidget {
  const HomeScreen({super.key});

  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen> {
  Color _gestureColor = const Color(0xFFE9EDFF);
  bool _isAlternateColor = false;

  void _showMessage(String message) {
    ScaffoldMessenger.of(context)
      ..hideCurrentSnackBar()
      ..showSnackBar(SnackBar(content: Text(message)));
  }

  void _changeColor() {
    setState(() {
      _isAlternateColor = !_isAlternateColor;
      _gestureColor = _isAlternateColor
          ? const Color(0xFFD8F5E7)
          : const Color(0xFFE9EDFF);
    });
    _showMessage('Single tap detected: color changed');
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Experiment 7'),
        centerTitle: true,
      ),
      body: SafeArea(
        child: ListView(
          padding: const EdgeInsets.all(20),
          children: [
            Text(
              'Navigation and Gestures',
              style: Theme.of(context).textTheme.headlineMedium?.copyWith(
                    fontWeight: FontWeight.bold,
                  ),
            ),
            const SizedBox(height: 8),
            Text(
              'Try basic navigation, named routes, and three common gestures.',
              style: Theme.of(context).textTheme.bodyLarge,
            ),
            const SizedBox(height: 24),
            Card(
              clipBehavior: Clip.antiAlias,
              child: Padding(
                padding: const EdgeInsets.all(20),
                child: Column(
                  children: [
                    GestureDetector(
                      key: const Key('namedRouteLogo'),
                      onTap: () => Navigator.pushNamed(context, '/second'),
```

```
                    child: const Semantics(
                      button: true,
                      label: 'Open second screen using named route',
                      child: FlutterLogo(size: 112),
                    ),
                  ),
                  const SizedBox(height: 12),
                  const Text(
                    'Tap the Flutter logo to open the second screen using a named route.',
                    textAlign: TextAlign.center,
                  ),
                ],
              ),
            ),
          ),
          const SizedBox(height: 16),
          GestureDetector(
            key: const Key('gestureArea'),
            onTap: _changeColor,
            onDoubleTap: () => _showMessage('Double tap detected'),
            onLongPress: () => _showMessage('Long press detected'),
            child: AnimatedContainer(
              duration: const Duration(milliseconds: 250),
              padding: const EdgeInsets.symmetric(
                horizontal: 20,
                vertical: 28,
              ),
              decoration: BoxDecoration(
                color: _gestureColor,
                borderRadius: BorderRadius.circular(20),
                border: Border.all(color: const Color(0xFFB9C4F8)),
              ),
              child: const Column(
                children: [
                  Icon(Icons.touch_app_rounded, size: 42),
                  SizedBox(height: 10),
                  Text(
                    'Gesture Playground',
                    style: TextStyle(
                      fontSize: 18,
                      fontWeight: FontWeight.bold,
                    ),
                  ),
                  SizedBox(height: 6),
                  Text(
                    'Tap to change color • Double tap for a message • Long press for a message',
                    textAlign: TextAlign.center,
                  ),
                ],
              ),
            ),
          ),
          const SizedBox(height: 24),
          FilledButton.icon(
            key: const Key('basicNavigationButton'),
            onPressed: () {
              Navigator.push(
                context,
                MaterialPageRoute(
                  builder: (context) => const SecondScreen(
                    openedWith: 'Navigator.push()',
                  ),
                ),
              );
            },
            icon: const Icon(Icons.arrow_forward_rounded),
            label: const Text('Open with Navigator.push'),
          ),
        ],
      ),
    ),
  );
  }
}
```

second screen dart

```
import 'package:flutter/material.dart';

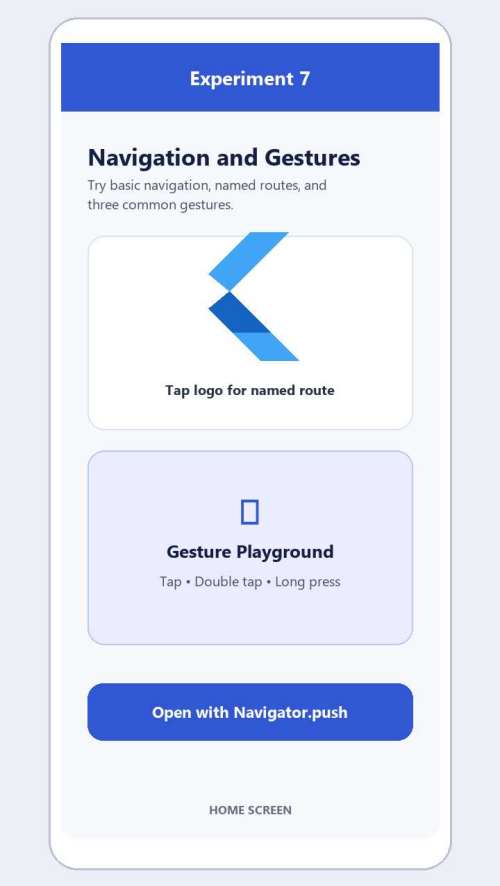
class SecondScreen extends StatelessWidget {
  const SecondScreen({
    super.key,
    required this.openedWith,
  });

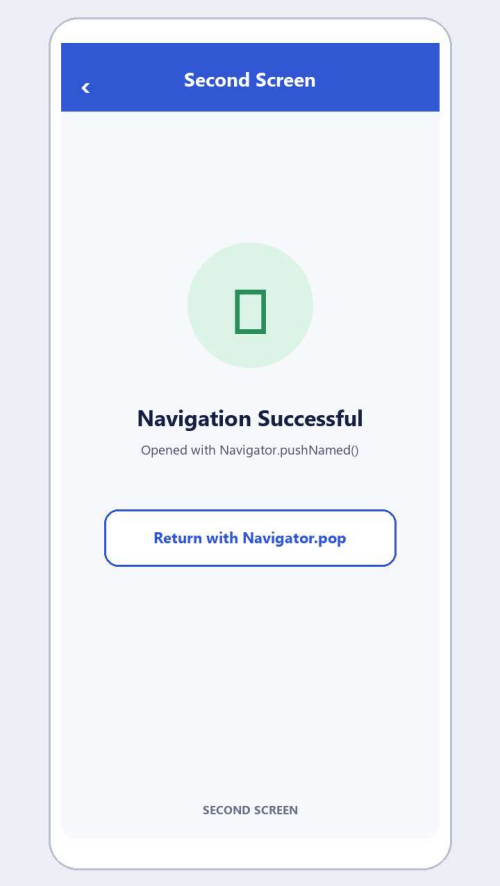
  final String openedWith;

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(title: const Text('Second Screen')),
      body: Center(
        child: Padding(
          padding: const EdgeInsets.all(24),
          child: Column(
            mainAxisSize: MainAxisSize.min,
            children: [
              const Icon(
                Icons.check_circle_rounded,
                size: 88,
                color: Color(0xFF2A8F5B),
              ),
              const SizedBox(height: 16),
              Text(
                'Navigation Successful',
                style: Theme.of(context).textTheme.headlineSmall?.copyWith(
                      fontWeight: FontWeight.bold,
                    ),
              ),
              const SizedBox(height: 8),
              Text(
                'Opened with $openedWith',
                textAlign: TextAlign.center,
              ),
              const SizedBox(height: 24),
              OutlinedButton.icon(
                key: const Key('backButton'),
                onPressed: () => Navigator.pop(context),
                icon: const Icon(Icons.arrow_back_rounded),
                label: const Text('Return with Navigator.pop'),
              ),
            ],
          ),
        ),
      ),
    );
  }
}
```

OUTPUT

The home screen provides two navigation methods and a gesture playground. The second screen confirms how it was opened and returns with Navigator.pop().





RESULT

A Flutter mini-application was implemented with basic navigation, named routing and gesture detection. The source opens and closes screens through Navigator and handles tap, double-tap and long-press input with visual feedback.

CONCLUSION

This experiment demonstrated how Flutter's navigation stack and named routes can organize movement between screens. GestureDetector was used to make widgets interactive, while SnackBar and color changes provided immediate feedback. Combining these features produced a structured and user-friendly mobile application aligned with CO 2 at the Apply level.